

Associations between tropical cyclone threat, impact, and recovery and adverse pregnancy
outcomes: A retrospective cohort study of Florida births

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This research was supported by the National Academies of Science Gulf Research Program
Understanding the Effects of Climate Change on Environmental Hazards in Overburdened
Communities (SCON-10000677).

Abstract

Background: Climate change is increasing the intensity of tropical cyclones (TCs), amplifying their environmental and public health impacts. Research examining TC exposure during pregnancy on adverse pregnancy outcomes has often used broad exposure definitions and examined a limited set of outcomes.

Objective: This study evaluated associations between TC exposure and adverse pregnancy outcomes (preterm birth (PTB), very preterm birth (vPTB), term low birthweight (tLBW), low birthweight (LBW), very low birthweight (vLBW), gestational diabetes (GDM), and hypertensive disorders of pregnancy (HDP)).

Methods: We constructed a retrospective cohort study using Florida birth certificate records from 2000-2022 (N = 4,514,346). TC data was imported from the National Oceanic and Atmospheric Administration (NOAA) IBTrACS dataset, which was spatially joined with maternal residence at birth to determine whether the births occurred within the TC's effect zone during each trimester.

Four binary exposure measures were created: threat, impact, cleanup, and any TC effect.

Logistic regression was used to estimate adjusted odds ratios (aOR).

Results: We found increased odds of PTB in the first (aOR 1.012, 95% CI 1.004, 1.020) and second trimester (aOR 1.008, 95% CI 1.000, 1.016) and increased odds of tLBW in the first (aOR 1.023, 95% CI 1.006, 1.040) and second trimester (aOR 1.034, 95% CI 1.019, 1.051).

Experiencing any TC effect was associated with increased odds of GDM (aOR 1.05, 95% CI 1.041, 1.068) and HDP (aOR 1.127, 95% CI 1.114, 1.141). Decreased odds were observed for vPTB and vLBW in the second trimester and tLBW in the third trimester. Similar results were observed when examining threat, impact, and cleanup TC effects.

Significance: These findings suggest that TC exposure during pregnancy may influence maternal and infant health. Improving exposure assessment and understanding mechanisms between TC exposure and adverse pregnancy outcomes are important for informing strategies to protect pregnant populations.

Keywords: tropical cyclones, preterm birth, low birthweight, gestational diabetes, climate change

Impact: Climate change is intensifying tropical cyclones, yet evidence on pregnancy health impact remains limited and often relies on broad exposure definitions. Using 4.5 million Florida births from 2000-2022 and spatially linking cyclone track data, this study improves exposure assessment by distinguishing threat, impact, and recovery phases across pregnancy trimesters.

Results show higher odds of preterm birth, term low birthweight, gestational diabetes, and hypertensive disorders of pregnancy following cyclone exposure. These findings highlight pregnancy as a sensitive period for climate-related hazards and support stronger disaster preparedness and public health planning to protect pregnant people and infants in cyclone-prone areas.

Introduction

Tropical cyclones (TC), severe storm systems classified by wind speeds such as tropical depressions, tropical storms, and hurricanes (also referred to as typhoons and cyclones in some regions), cause not only severe physical damage to coastlines, infrastructure, and homes, but also have detrimental effects on human health. It is estimated they have cost the United States nearly \$1 trillion since 1980 and have resulted in around 50 direct deaths each year¹. With climate change, TCs are becoming more intense², increasing the environmental and public health impacts they have on Gulf Coast communities (e.g. Florida, Texas, Louisiana, Alabama, Mississippi). Gulf Coast communities may particularly suffer from climate-related hazards as a result of their concentration of many socially disadvantaged populations³. From 2010 to 2020, Florida has seen an 8.2% and 34.9% increase in Black and Hispanic populations, respectively⁴. Minority individuals are often disproportionately affected by hurricane-related impairments such as PTSD, disability, and job loss, which may constrain their access to healthcare and increase the risk of adverse health outcomes post-hurricane⁵⁻⁷.

Pregnancy is a period of heightened physiological and psychological vulnerability, during which adverse maternal and infant health may be particularly exacerbated by environmental stressors. Several studies have shown positive associations between natural disasters and prenatal maternal stress, and maternal stress during pregnancy from natural disasters may impact birth outcomes and child development⁸⁻¹⁰. For example, a meta-analytical review indicated that natural disaster-related prenatal maternal stress significantly correlated with adverse birth outcomes ($r = 0.0547$, where r represents correlation coefficient; 95% CI 0.0256, 0.0836)¹¹. As a significant disaster, TC exposure would be hypothesized to cause adverse pregnancy outcomes in part by increasing prenatal maternal stress. However, research evaluating associations between TC

exposure during pregnancy and adverse pregnancy outcomes is somewhat mixed. Antipova et al., for example, found preterm deliveries were more likely following Hurricane Andrew compared to pre-event births (OR 1.43, 95% CI 1.26, 1.62),¹² while Liu et al. observed no association between exposure to Hurricane Harvey and spontaneous preterm birth (PTB) (OR 1.06, 95% CI 0.91, 1.25)¹³. Notably, a recent meta-analysis similarly found no overall association between storm exposure and preterm birth or low birthweight (LBW)¹⁴. Additionally, most prior studies have primarily focused on adverse birth outcomes such as preterm birth and low birthweight, with limited attention on pregnancy complications including gestational diabetes (GDM) and hypertensive disorders of pregnancy (HDP).

Given the lack of consistency in these findings and limited evaluation of maternal complications, we examined associations between TC exposure in Florida and adverse pregnancy outcomes (e.g. PTB, very preterm birth (vPTB), term low birthweight (tLBW), LBW, very low birthweight (vLBW), GDM, and HDP). To obtain a more expansive characterization of TC exposure, we evaluated different aspects of hurricane experiences, including hurricane threat, impact, and recovery (cleanup days). Moreover, rather than focus on a single TC event, we synthetically looked across 27 tropical cyclones in Florida, U.S. to assess generalizable effects of TC exposure on adverse pregnancy outcomes. We hypothesized that experiencing a TC is associated with increased odds of adverse pregnancy outcomes.

Materials and Methods

Study Population

We conducted a retrospective cohort study of mothers who gave birth in Florida between 2000 and 2022 using birth records from the Florida Department of Health. Our analysis was

restricted to observations with accurate latitude and longitude (N = 4,662,195) and singleton births (N = 4,514,346) during the study period, and we excluded births with missing data on gestational weeks and birthweight (N = 279). We restricted our tLBW analysis to term births (N = 4,125,271). Furthermore, in our analysis of gestational hypertensive disorders, we excluded mothers with missing data on GDM and HDP (N = 664,775), as well as mothers with chronic hypertension (N = 31,983) or diabetes (N = 56,253), leaving 3,761,335 observations for the analysis of pregnancy complications.

Outcome

The outcomes of interest were PTB, vPTB, tLBW, LBW, vLBW, HDP, and GDM. PTB was defined as gestational weeks less than 37 weeks. vPTB was defined as gestational weeks less than 32 weeks. tLBW was defined as a birthweight less than 2500 grams at 37 or more weeks. LBW was defined as birthweight less than 2500 grams. vLBW was defined as birthweight less than 1500 grams. GDM was defined as mothers diagnosed with gestational diabetes during pregnancy. HDP were defined as mothers diagnosed with gestational hypertension (e.g. pregnancy-induced hypertension, preeclampsia, etc.) during pregnancy. Gestational weeks, birthweight, GDM, and HDP were obtained from the Florida Department of Health Vital Statistics birth records.

Exposure

TC data was imported from the North Atlantic tropical cyclone track data from the National Oceanic and Atmospheric Administration (NOAA) IBTrACS dataset which includes storm track geometries and attributes like wind speed and pressure. Tracks were read and

transformed to EPSG:32616 (UTM Zone 16N) for buffering. We filtered tracks to include significant storms (e.g. TC with wind speeds ≥ 28 m/s [55 knots]) that occurred between 2000 and 2022. We buffered each TC's track using the radius to maximum winds (RMW) to create polygonal wind swaths to represent TC size. Missing RMW values were imputed through within-storm mean, then by mean within the same storm central pressure stratum, and re-checked within storm. The swaths were spatially intersected with a boundary defining the state of Florida to identify TCs affecting the region. Overlapping swaths from the same TC were dissolved to form one geometry per TC and assigned a Saffir-Simpson category based on peak wind speed. Swath geometries were then transformed into a geographic coordinate system for compatibility with later spatial operations. We expanded the modeled TC footprints into a time series that included the day before landfall, the day of landfall, and the day after landfall and used them to classify exposure windows.

Individual-level birth data were joined with the TC data to assess exposure during each trimester. We filtered TCs to those falling within an individual's trimester of pregnancy. Spatial join was used to determine whether the birth location falls within the TC's effect zone during each trimester. Those without records of TC overlap were considered non-exposed.

With the linked TC and birth data, we created four binary exposure variables: threat, impact, cleanup, and any TC effect. Threat was defined as those who experienced the day before landfall during the trimester. Impact was defined as those who experienced the day of landfall during the trimester. Cleanup was defined as those who experienced the day after landfall during the trimester. Any TC effect was defined as those who experienced either threat, impact, or cleanup. Each of the four variables was created for each trimester.

Covariates

Covariates and potential confounders were determined a priori based on their relation to TC exposure or the adverse pregnancy outcomes. As TC exposure is strongly influenced by place-based characteristics, associations between TC exposure and pregnancy complications may reflect underlying geographic differences in demographic and socioeconomic characteristics of affected communities rather than the storm exposure itself. Maternal and paternal characteristics (mother's age, mother and father's race/ethnicity, mother and father's education, pre-pregnancy body mass index (BMI), previous preterm infant, alcohol use, principal source of payment, and Women, Infant, and Child (WIC) program status) were ascertained from birth records. We categorized pre-pregnancy BMI into the follow categories: underweight (BMI < 18.5), normal weight (BMI 18.5 – 24.9), overweight (BMI 25 – 29.9), and obesity (BMI > 30).

Statistical Analysis

Demographic and categorical differences in the study population were assessed using summary statistics. To assess the association between TC effect and PTB, vPTB, tLBW, LBW, and vLBW, we used logistic regression models and calculated adjusted odds ratios (aOR) and 95% confidence intervals (CI) during each trimester. As those born preterm do not have a complete third trimester, results for PTB, vPTB, LBW, and vLBW were not estimated for this trimester. To assess the association between TC effect and GDM and HDP, we used logistic regression models and calculated aOR and 95% CI in the first trimester exposure. We focused on first trimester exposure as GDM and HDP are tested for after 20 and 24 weeks of pregnancy, respectively, allowing sufficient time for these conditions to develop following exposure. To control for confounding, models were adjusted for the mother and father's race/ethnicity and

mother and father's education. Race/ethnicity and education may correspond to neighborhood development and may influence where TC exposure is experienced, as well as an increased risk of pregnancy complications and adverse birth outcomes. Models were also adjusted for the following covariates as they are related to the risk of pregnancy complications and adverse birth outcomes can increase precision of the estimates: mother's age, pre-pregnancy BMI, previous preterm infant, WIC status, and principal source of payment. To handle missing data among covariates (Table 1), multiple imputation by chained equations was used to impute null values with five iterations; missingness was highest for mother (22.2%) and father's (33.6%) race, father's ethnicity (27.6%), and pre-pregnancy BMI (20%). Multiple imputation is a statistical approach that fits a regression model for each covariate with missing values, conditioning on all other covariates. Missing values are then imputed sequentially and repeated for five iterations to obtain stable results¹⁵. All analyses were conducted in R version 4.5.2.

Results

Of the 4.5 million births in our study, 8.6% were delivered preterm, 1.2% were delivered very preterm, 6.8% had LBW, and 1.2% had vLBW. When the study population was restricted to term births, 2.5% had LBW. 4.9% and 5.7% of mothers had GDM and HDP, respectively. Our total analytic sample was majority White (55.8%) and non-Hispanic (56.4%). On average, women were 28.2 years old at delivery. Among the study population, 31.3% have a high school diploma or GED, 37.6% had a normal pre-pregnancy weight, and 39.4% and 36.4% had Medicaid or private insurance, respectively (Table 1). Overall, 80.8% of mothers experienced some effect of TC exposure.

Any TC effect was associated with increased odds of PTB in the first (aOR 1.012, 95% CI 1.004, 1.020) and second trimester (aOR 1.008, 95% CI 1.000, 1.016) (Table 2). Decreased odds were observed for VPTB in the second trimester (aOR 0.919, 95% CI 0.902, 0.937). Similar results were seen when examining threat, impact, and cleanup TC effects.

We observed increased odds of tLBW in the first (aOR 1.023, 95% CI 1.006, 1.040) and second trimester (aOR 1.034, 95% CI 1.019, 1.051) (Table 3). Any TC effect was associated with decreased odds of vLBW (aOR 0.914, 95% CI 0.896, 0.933) in the second trimester and tLBW in the third trimester (aOR 0.966, 95% CI 0.950, 0.982). These results were also seen when examining threat, impact, and cleanup TC effects.

Experiencing any TC effect was associated with increased odds of GDM (aOR 1.05, 95% CI 1.041, 1.068) and HDP (aOR 1.127, 95% CI 1.114, 1.141) (Table 4). When examining threat, impact, and cleanup TC effects, results were similar to that of any TC effect.

Discussion

We examined associations between TC exposure in Florida and adverse pregnancy outcomes (PTB, vPTB, tLBW, LBW, vLBW, GDM, and HDP). We found increased odds of PTB and tLBW with first and second trimester TC exposure. In the third trimester, decreased odds of LBW and tLBW were observed. Second trimester TC exposure resulted in decreased odds of vPTB and vLBW. Furthermore, we observed increased odds of GDM and HDP with first trimester TC exposure. Characterization of TC exposure (threat, impact, cleanup) did not make a substantial difference on the effect of TC exposure on pregnancy outcomes.

The decreased odds observed in the second trimester for vPTB and vLBW and the third trimester for tLBW should be interpreted cautiously. One possible explanation is that individuals

later in pregnancy may have altered behaviors in anticipation of TC events, which could influence the timing of birth outcomes. It is also possible that timing of TC exposure affects biological pathways differentially, and early gestation may represent a more sensitive period of stress-related placental disruption compared to later in pregnancy.

Our findings are both consistent and divergent with previous research. Multiple studies assessing this association have provided mixed findings with positive^{12,16-19 23}, negative²⁰, and null associations^{13-14, 21-23}. With regards to PTB, our results are consistent with a previous study of tropical cyclones in 378 eastern US counties and found a higher risk of preterm birth with tropical cyclone exposure defined as peak wind speed >17.2 m/s (RR 1.01, 95% CI 0.99, 1.03) and distance to storm track < 60 km (RR 1.02, 95% CI 1.01, 1.04)¹⁸. The authors compare areas affected and unaffected by tropical cyclones in early pregnancy and found higher odds of preterm birth (OR 1.28, 95% CI 1.11, 1.49). For vPTB, our results are inconsistent with a 2016 Florida study examining Hurricane Charley exposure which found an increased hazard of extremely preterm delivery (HR 1.09, 95% CI 1.03, 1.16)²⁴. A 2023 meta-analysis analyzing the health effects of cyclones did not find a significant association between cyclones and preterm birth (meta-RR 1.09, 95% CI 0.91, 1.29)²⁵. Our findings indicating increased odds of HDP and GDM are also consistent with emerging evidence linking TC exposure to adverse maternal cardiometabolic outcomes; a 2019 study found increased odds of hypertensive disorders of pregnancy following landfall of Hurricane Harvey in Texas (aOR 1.60, 95% CI 1.37, 1.86)²⁶.

The results from these studies as well as the conflicting findings could be due to differences in study design. One methodological difference that may be contributing to the inconsistent finding for this association is how TCs are defined and operationalized across studies. Exposure definitions in prior research vary widely. Some rely on a subject's perceived

experience of TC exposure or residing in a disaster declaration area, while others use measures such as maximum wind speed and county-level exposure. Studies using county-level exposure or disaster declaration areas may be prone to bias from exposure misclassification as they often assume a uniform storm impact within these geographic areas, even though there may be differences in infrastructure, evacuation behavior, wind, and rainfall²⁷⁻²⁸. Additionally, studies differ in the temporal windows used to assign exposure, with some defining exposure by date of landfall, others examining broader periods like weeks or months before or after the TC. The variation in temporal windows may contribute to the inconsistency in results. For example, a study examining the risk of spontaneous early-term birth in relation to Hurricane Harvey found an increase in the odds of early term birth within one week of landfall (OR 1.14, 95% 1.04, 1.25), but the results were null when considering 2-week and 4-week lags¹³. These variations in temporal windows and exposure classifications make it challenging to compare findings across studies. In contrast, our study evaluated multiple dimensions of TC exposure, including storm threat, impact, and cleanup periods. While findings were generally consistent across these operationalizations and exact timing didn't make a difference in results, our study used a more precise definition of direct exposure to the storm (a mother's residential address being within a certain distance of the storm), which to our knowledge most similar studies have not used.

This study has a few limitations. Even with detailed storm-track modeling, subjects did not all experience TC in the same way, so our measure of exposure is limited. Using birth residence to define exposure may not accurately reflect where a mother resided during a storm; therefore, exposure misclassification may exist. Additionally, TC-related displacement or long-term relocation could introduce selection bias if individuals who remained in affected areas and delivered there differ systematically in socioeconomic resources, vulnerability, or health status

from those who left after a TC. We were not able to account for whether pregnant individuals evacuated, sheltered in place, or were outside of the TC track during the storm. By using binary exposure variables (i.e., any TC, threat, impact, cleanup), we may miss differences in storm severity, flooding, or personal disruption; moreover, these exposure categories often co-occur, particularly when TCs occur near trimester boundaries, meaning individuals who experienced one component often experienced all of them. Furthermore, we are limited to the information presented in the birth records. Prior studies have documented birth certificates contain inaccuracies for HDP and GDM, which are underreported and misclassified compared to medical records²⁹⁻³¹. Such misclassification may potentially bias our results towards the null if nondifferential; however, differential misclassification could have occurred if reporting practices differed in storm affected and unaffected areas (i.e., due to healthcare system disruptions), potentially biasing results in unexpected directions. The birth certificates used in the present study also lack information on important factors such as prenatal maternal stress, evacuation behavior, housing and neighborhood conditions after the storm, and prenatal care disruptions during the storm that may confound or mediate the association between hurricane exposure and adverse pregnancy outcomes. Finally, our study used data from Florida only and therefore may not fully generalize to other Gulf Coast states or hurricane-prone regions. Florida's disaster preparedness and resident's resilience may differ from other states and regions. For example, compared to counties in Louisiana, Mississippi, and Alabama, coastal counties in Florida exhibited higher levels of resilience to climate and natural hazard stressors due to higher levels of governance and stronger social systems³². Moreover, although Florida has an extensive maternal and child health infrastructure, including obstetric providers, public health agencies, and community-based services, there are still barriers to disaster planning, communication, and

284 post-disaster recovery that hinder the ability to fully address the needs of pregnant women during
285 natural disasters³³.

286 TC exposure during pregnancy was associated with increased odds of preterm birth, low
287 birthweight, hypertensive disorders of pregnancy, and gestational diabetes. These findings add to
288 a growing body of evidence suggesting that extreme weather events may adversely affect both
289 maternal and infant health. As the frequency and intensity of TCs are projected to increase with
290 climate change, improving exposure assessment and understanding mechanisms such as prenatal
291 maternal stress and disruptions in prenatal care will be important for informing strategies to
292 protect pregnant individuals and their infants.

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296 Data Availability: The datasets generated during and/or analyzed during the current study are not
297 publicly available for protection of subject privacy and due to the state of Florida's terms for
298 accessing the data.

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411 Acknowledgements: Not applicable

412 Author Contributions: JAB and EWH conceptualized the study. JAB analyzed and interpreted
413 the data and drafted the manuscript. EWH interpreted the data. JE, DH, and CKU acquired and
414 created the data. EWH, JE, DH, and CKU substantially revised the manuscript. All authors read
415 and approved the final manuscript.

416 Funding: This research was supported by the National Academies of Science Gulf Research
417 Program Understanding the Effects of Climate Change on Environmental Hazards in
418 Overburdened Communities (SCON-10000677). Any published findings and conclusions are
419 those of the authors and do not necessarily represent the official position of the Florida
420 Department of Health.

421 Ethical Approval: These analyses were conducted under a waiver of informed consent and
422 approval by the Institutional Review Boards of Tulane University, Florida State University
423 #5297, and the Florida Department of Health #2018-066-FSU.

424 Competing interests: The authors declare that we have no competing interest.

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